analyzer, the same analyzer is used during the search phase. This guarantees that words indexed and inserted into inverted indexes are also matched while searching.

As in any application, a bunch of building blocks make up the Elasticsearch server. We discuss those in the next section, and we work extensively with them throughout this book.

3.2 The building blocks

In the last chapter, we indexed sample documents and carried out searches on them. However, we pushed the discussion of components like indexes, documents, shards, and replicas to a later time—and that time is here. These components make up Elasticsearch and are the building blocks of the search engine. Let's examine them in detail and learn their significance.

3.2.1 Documents

A *document* is the basic unit of information indexed by Elasticsearch for storage. Each document's individual fields are analyzed for faster searches and analytics. Elasticsearch expects documents in JSON format. JSON is a simple human-readable data format that has gained popularity in recent years. It represents data as key-value pairs: for example `{"name":"John Doe"}`. When we communicate with Elasticsearch over RESTful APIs, we send queries to Elasticsearch as JSON objects. Elasticsearch, in turn, serializes these JSON documents and stores them in its distributed document store once they are analyzed.

PARSING DOCUMENT DATA

Our data, represented as JSON documents, is parsed by Elasticsearch during the indexing process. For example, figure 3.3 represents a JSON document for a `student` object.

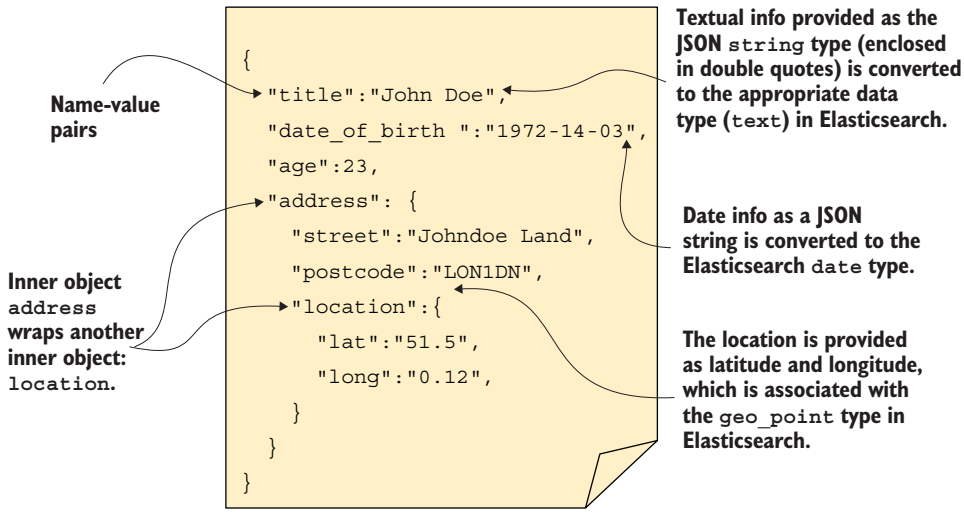


Figure 3.3 A student document represented in JSON format